Dual-Flux Systems Biology Lab

A conceptual open research repository dedicated to systems-level biological energetics, regenerative stability, transport-limited growth, and interdisciplinary models of living systems.

This repository develops the idea that multicellular organisms operate through partially interconnected energetic regimes rather than through a single universal metabolic mode. The central framework is the Dual-Flux Energy Architecture model, which describes biological systems through the interaction between:

- Oxidative Mitochondrial Functional Flux
  - high-power functional metabolism,
  - mature tissue activity,
  - oxidative phosphorylation,
  - long-term infrastructure wear.

- Basal Glycolytic Regime (BGR)
  - glycolytic-recycling metabolism,
  - glucose-lactate buffering networks,
  - regenerative stability,
  - stem-cell maintenance,
  - reproductive biology,
  - sleep-associated repair,
  - and protection of biological information integrity.

The project aims to create a common language connecting:

- biology,
- physics,
- chemistry,
- mathematics,
- and systems medicine.

Main Research Directions

1. Dual-Flux Energy Architecture
A systems-level framework describing biological energetics as the interaction of functional oxidative metabolism and protective glycolytic-recycling regimes.

2. Basal Glycolytic Regime (BGR)
A proposed protective metabolic mode associated with:
- reduced oxidative stress,
- distributed glucose-lactate recycling,
- regeneration,
- developmental biology,

- and long-term biological stability.

3. Transport-Limited Growth
Growth as a process constrained not only by energy availability, but by:
- vascular architecture,
- substrate transport,
- stable flux maintenance,
- and infrastructure cost.

Applications include:
- pregnancy physiology,
- regenerative biology,
- tissue engineering,
- and tumor energetics.

4. Aging as Energetic Redistribution
Aging interpreted as:
- increasing infrastructure maintenance cost,
- progressive deformation of energetic stability landscapes,
- declining regenerative reserve,
- and reduced access to protective metabolic regimes.

5. Sleep as Infrastructure Maintenance
Sleep explored as:
- a cyclic repair state,
- mitochondrial quality-control window,
- and systemic restoration phase of biological energy networks.

6. Regeneration Pendulum
A dynamic model of tissue repair involving oscillatory interactions between:
- proliferation,
- energy deficit,
- apoptosis,
- matrix remodeling,
- vascular adaptation,
- and restoration.

7. Physical Models of Cellular Stability
Conceptual models describing the cell as:
- a confined hydrogel system,
- a non-equilibrium dissipative structure,
- and a transport-constrained energetic network.

Repository Structure

/foundations

Core conceptual framework papers and theoretical architecture.

/biology
Developmental biology, stem cells, regeneration, pregnancy, aging, and tumor energetics.

/physics
Thermodynamics, dissipative structures, transport systems, and energetic stability models.

/chemistry
Redox architecture, lactate transport, substrate cycling, and metabolic buffering systems.

/mathematics
Dynamical systems, attractor landscapes, bifurcation models, and flux equations.

/clinical_models
Applications to medicine, sleep, metabolic disease, aging, regenerative medicine, and systems pathology.

/drafts
Work-in-progress conceptual papers and exploratory models.

Philosophy of the Repository

This repository is not intended as a closed doctrine or finalized theory.

It is an open conceptual laboratory intended for:
- interdisciplinary discussion,
- theoretical exploration,
- mathematical formalization,
- biological validation,
- and collaborative systems thinking.

The central premise is simple:

> Living systems may be understood more clearly when biology, physics, chemistry, and mathematics describe the same energetic object through a shared language.

Current Status

This repository contains:

- conceptual framework papers,
- theoretical essays,
- exploratory systems models,
- and interdisciplinary hypotheses.

Many models remain preliminary and require:
- mathematical formalization,
- experimental testing,
- biological validation,
- and critical analysis.

Contributions, critiques, alternative formulations, and formal models are welcome.
